What is the target variable and why?

The target variable for our project is ‘life expectancy at birth’. According to the World Bank, life expectancy at birth represents the average number of years a newborn is expected to live if mortality patterns at the time of birth remain constant in the future. This measure reflects the overall mortality level of a population and summarizes the mortality patterns across all age groups in a given year. It is calculated using a period life table, which provides a snapshot of a population’s mortality at a specific time.

Life expectancy is a powerful indicator of public health and development. It is widely used in global health research because it provides a broad and meaningful measure of population health across countries. High mortality in younger age groups significantly lowers life expectancy at birth. If an individual survives childhood, they are likely to live much longer. We believe that this will be an essential metric for understanding the impact of healthcare disparities on long-term health outcomes.

This variable is particularly useful for comparing different countries and income levels. It reflects key factors such as healthcare access, economic stability, and public health policies. Because reliable data on disease prevalence is often unavailable, mortality rates including life expectancy are frequently used to assess healthcare effectiveness. As one of the most important indicators for comparing socioeconomic development across countries, life expectancy at birth serves as a comprehensive variable for analyzing our problem statement on healthcare inequities over time.

What are the predictors and why?

To analyze life expectancy, we included predictors related to healthcare resources, health outcomes, expenditures, socioeconomics, public health, and environmental factors.

Healthcare resources include the availability of physicians, nurses, and hospital beds. A well-equipped healthcare system is essential for improving life expectancy. Countries with more medical staff and infrastructure generally provide better healthcare services, leading to longer lives. We also include specialist surgical workforce data and the percentage of births attended by skilled health staff as these indicators reflect access to critical healthcare services.

Health outcomes such as maternal and infant mortality reflect the effectiveness of a healthcare system. High mortality rates often signal gaps in care, especially in lower-income countries.

Mortality from chronic diseases like cancer, cardiovascular disease, and diabetes also impact life expectancy. The prevalence of preventable deaths emphasizes the need for better healthcare access and treatment options.

Healthcare expenditures help assess the financial accessibility of healthcare. We included total health expenditure as a percentage of GDP, government and private health spending, and out-of-pocket expenses. The proportion of people spending a significant share of their income on healthcare provides a measure of financial burden. High out-of-pocket costs can limit access to essential services, thereby lowering life expectancy.

Socioeconomic factors such as gross national income, poverty levels, and human capital index also contribute to disparities in healthcare access. Higher-income countries tend to have better healthcare services and longer life expectancy. Poverty headcount ratios and urban poverty levels highlight economic inequalities. These factors will help to explain variations in health outcomes.

Public health metrics will provide additional context. Immunization coverage, access to clean drinking water, and sanitation services affect disease prevention and overall health. Countries with strong public health policies often experience lower mortality rates and longer life expectancy. These variables help connect healthcare access with broader social determinants of health.

Exploration of the dataset

Our dataset contains 1,655 rows and 24 columns, representing various healthcare-related indicators across multiple countries. Each row corresponds to a unique observation of a specific variable for a given country and year. Our dataset includes a mix of categorical and numerical variables:

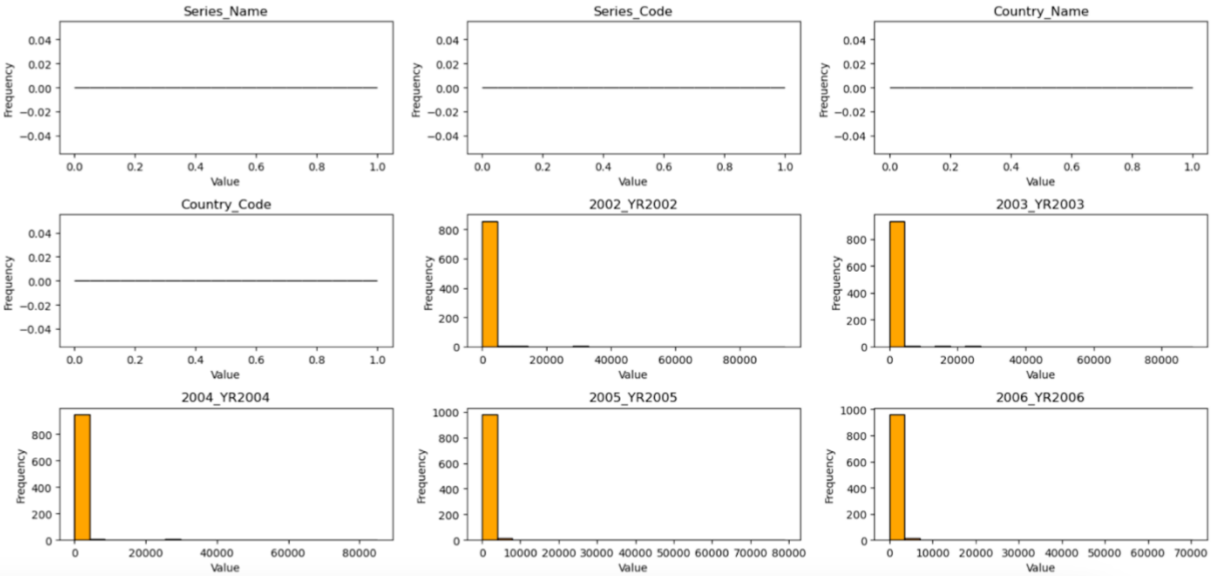
- Categorical variables:
    - Series Name – describes the indicator being measured.
    - Series Code – a unique identifier for each indicator.
    - Country Name and Country Code – represent each country in the dataset.
    - Category – classifies indicators into groups (e.g., Health Outcomes, Healthcare Expenditure).
    - Income Group – categorizes countries into high, middle, and low-Income groups.

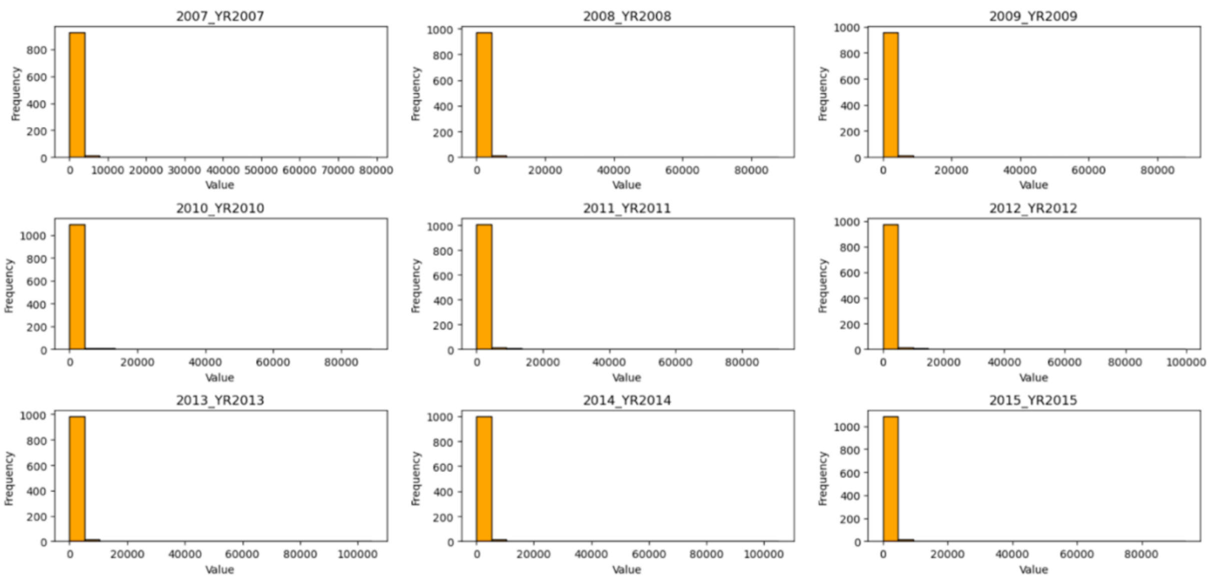
- Numerical variables:
  - These are yearly data columns, capturing time-series trends of various healthcare and socio-economic indicators from 2002 to 2021.

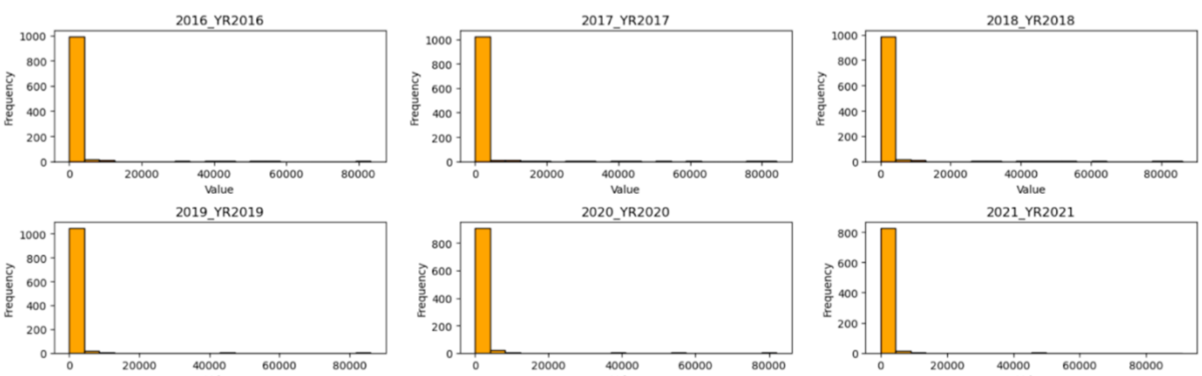
For a focused analysis, we selected the most important predictors from each category, including:

- Healthcare Resources:
  - Physicians (per 1,000 people) – Indicator of healthcare accessibility and quality.
  - Hospital beds (per 1,000 people) – Measures inpatient care capacity.
- Healthcare Outcomes:
  - Maternal mortality ratio (per 100,000 live births) – Reflects maternal and prenatal care quality.
  - Mortality rate, under-5 (per 1,000) – Measures child health and healthcare effectiveness.
  - Incidence of Tuberculosis (per 100,000 people) – Represents the burden of infectious diseases.
- Healthcare Expenditures:
  - Current health expenditure per capita (current US$) – Directly influences healthcare quality and outcomes.
  - Out-of-pocket expenditure (% of current health expenditure) – Indicates financial burden on individuals.
- Demographics & Socioeconomics:
  - Population ages 65 and above (% of total population) – Affects healthcare demand and costs.
- Public Health Metrics:
  - UHC service coverage index – Measures access to essential health services.
- Water, Sanitation, and Hygiene (WASH):
  - People using safely managed drinking water services (% of population)– Directly impacts disease prevalence and public health.

Exploring Data Distributions







The analysis reveals several key findings. Categorical variables like `Series_Name` and `Country_Code` show flat distributions, indicating they are likely categorical data or were improperly plotted as numeric values. The yearly data (2002–2021) is highly skewed, with most values concentrated near zero and a few extreme outliers. A significant portion of the dataset consists of zero or near-zero values, suggesting potential sparsity or missing data issues. Additionally, a few high-value outliers stand out, which may require further investigation. Notably, these patterns—skewness, sparsity, and outliers—are consistent across all years, suggesting systematic reporting trends. Moving forward, it would be important to investigate the outliers, address the skewness through data transformations, and assess the quality of the data.